Solutions to Miscellaneous Math Problems

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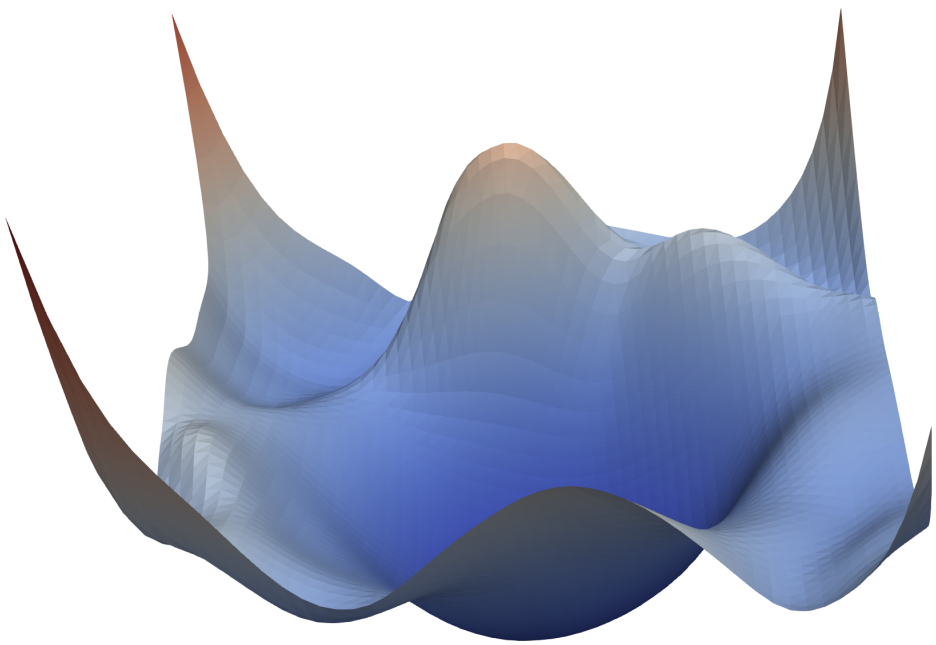
April 11, 2026

Abstract

This note collects my solutions to selected problems. The goal is to present clear arguments and organized solutions for study and review. Some topics in this note are: Machine Learning, Optimization, Statistical Learning Theory, etc.



Problem 1 (Natural Gradient).

We consider the problem of fitting model parameters using gradient descent. Fitting model parameters using Maximum Likelihood is equivalent to minimizing the KL divergence between the data and the model. A nice property of KL divergence is that it is invariant to parametrization. However, the gradient w.r.t the model parameters (i.e, direction of steepest descent) is *not invariant to parametrization.*

This is where the **natural gradient** comes into picture. It is best introduced in contrast with the usual gradient descent. In the usual gradient descent, we first choose the direction by calculating the gradient of the MLE objective w.r.t the parameters, and then move a magnitude of step size (where size is measured in the parameter space) along that direction. Whereas in natural gradient, we first choose a divergence amount by which we would like to move, in the D_{KL} sense. This effectively gives us a perimeter around the current parameters (of some arbitrary shape), such that points along this perimeter correspond to distributions which are at an equal D_{KL}-distance away from the current parameter. Among the set of all distributions along this perimeter, we move to the distribution that maximizes the objective (i.e minimize D_{KL} between data and itself) the most. This approach makes the optimization process invariant to parametrization. That means, even if we chose a new arbitrary reparametrization, by starting from a particular distribution, we always descend down the same sequence of distributions towards the optimum.

(a) Derive the natural gradient update rule.

(b) Show natural gradient based optimization is actually equivalent to Newton's method for Generalized Linear Models.

Source: Modified from CS 229 Stanford 2018 - Problem set 3

Solution:

(a) The natural gradient update rule is:

$$d^* = \arg\max_d \ell(\theta + d) \quad \text{subject to} \quad D_{\mathrm{KL}}(P_\theta || P_{\theta+d}) = c \quad (1)$$

Lemma 1: Approximating D_{KL} with Fisher Information

$$D_{\mathrm{KL}}(P_\theta || P_{\theta+d}) \approx \frac{1}{2} d^T \mathcal{I}(\theta) d$$

Consider the Taylor approximation of $\ell(\theta + d)$ around θ:

$$\ell(\theta + d) \approx \ell(\theta) + d^T \nabla_{\theta'} \ell(\theta') \Big|_{\theta'=\theta}$$

Plugging this approximation into (1), we get the following optimization problem:

$$d^* = \arg\max_d \ell(\theta) + d^T \nabla_{\theta'} \ell(\theta')\Big|_{\theta'=\theta} \quad \text{s.t.} \quad \frac{1}{2} d^T \mathcal{I}(\theta) d = c$$

We optimize the Lagrangian $\mathcal{L}(d, \lambda)$:

$$\mathcal{L}(d, \lambda) = \ell(\theta) + d^T \nabla_{\theta'} \ell(\theta')\Big|_{\theta'=\theta} - \lambda \left[\frac{1}{2} d^T \mathcal{I}(\theta) d - c\right]$$

- $\nabla_d \mathcal{L}(d, \lambda) = 0 \implies \nabla_{\theta'} \ell(\theta')\Big|_{\theta'=\theta} - \lambda \mathcal{I}(\theta) d = 0 \implies d = \frac{1}{\lambda} \mathcal{I}^{-1}(\theta) \nabla_{\theta'} \ell(\theta')\Big|_{\theta'=\theta}$
- $\nabla_\lambda \mathcal{L}(d, \lambda) = 0 \implies \frac{1}{2} d^T \mathcal{I}(\theta) d = c \implies d^T \mathcal{I}(\theta) d = 2c$

$$\begin{aligned}
&\implies d^T \nabla_{\theta'} \ell(\theta')\Big|_{\theta'=\theta} = 2c \implies \frac{1}{\lambda} \left(\nabla_{\theta'} \ell(\theta')\Big|_{\theta'=\theta}\right)^T \mathcal{I}^{-1}(\theta) \nabla_{\theta'} \ell(\theta')\Big|_{\theta'=\theta} = 2c \\
&\implies \lambda = \frac{1}{\sqrt{2c}} \sqrt{\left(\nabla_{\theta'} \ell(\theta')\Big|_{\theta'=\theta}\right)^T \mathcal{I}^{-1}(\theta) \nabla_{\theta'} \ell(\theta')\Big|_{\theta'=\theta}} \\
&\implies d^* = \frac{\sqrt{2c} \mathcal{I}^{-1}(\theta) \nabla_{\theta'} \ell(\theta')\Big|_{\theta'=\theta}}{\sqrt{\left(\nabla_{\theta'} \ell(\theta')\Big|_{\theta'=\theta}\right)^T \mathcal{I}^{-1}(\theta) \nabla_{\theta'} \ell(\theta')\Big|_{\theta'=\theta}}}
\end{aligned}$$

Proof of Lemma 1:

The KL divergence can be split into two expectations:

$$D_{\mathrm{KL}}(P_\theta || P_{\theta+d}) = \underbrace{\mathbb{E}_{x \sim P_\theta(x)}[\log P_\theta(x)]}_{A} - \underbrace{\mathbb{E}_{x \sim P_\theta(x)}[\log P_{\theta+d}(x)]}_{B}$$

Treating θ as a fixed point, A is constant. We perform a second-order Taylor Expansion on B around θ. Let $\tilde{\theta} = \theta + d$:

$$\begin{aligned}
B &= \mathbb{E}_{x \sim P_\theta(x)}[\log P_{\tilde{\theta}}(x)] \\
&\approx A + d^T \underbrace{\mathbb{E}_{x \sim P_\theta(x)} \left[\nabla_{\theta'} \log P_{\theta'}(x)\Big|_{\theta'=\theta}\right]}_{\text{First-order term}} + \frac{1}{2} d^T \underbrace{\mathbb{E}_{x \sim P_\theta(x)} \left[\nabla^2_{\theta'} \log P_{\theta'}(x)\Big|_{\theta'=\theta}\right]}_{\text{Second-order term}} d
\end{aligned}$$

Step 1: Show the first-order term is zero.

$$\begin{aligned}
\mathbb{E}_{x \sim P_\theta(x)} \left[\nabla_{\theta'} \log P_{\theta'}(x)\Big|_{\theta'=\theta}\right] &= \mathbb{E}_{x \sim P_\theta(x)} \left[\frac{\nabla_{\theta'} P_{\theta'}(x)}{P_{\theta'}(x)}\Bigg|_{\theta'=\theta}\right] \\
&= \int_{-\infty}^{\infty} P_\theta(x) \frac{\nabla_{\theta'} P_{\theta'}(x)\Big|_{\theta'=\theta}}{P_\theta(x)} dx \\
&= \nabla_{\theta'} \left[\int_{-\infty}^{\infty} P_{\theta'}(x) dx\right]\Bigg|_{\theta'=\theta} = \nabla_{\theta'}[1] = 0 \quad (*)
\end{aligned}$$

Step 2: Connect the second-order term to the Fisher Information Matrix $\mathcal{I}(\theta)$. By definition, and using the result from Step 1 $(*)$:

$$\begin{aligned}\mathcal{I}(\theta) &= \mathrm{Cov}_{x\sim P_\theta(x)}\left[\nabla_{\theta'}\log P_{\theta'}(x)\Big|_{\theta'=\theta}\right] \\ &= \mathbb{E}_{x\sim P_\theta(x)}\left[\left(\nabla_{\theta'}\log P_{\theta'}(x)\right)\left(\nabla_{\theta'}\log P_{\theta'}(x)\right)^T\Big|_{\theta'=\theta}\right]\end{aligned}$$

Examining a single element (i,j) of this matrix:

$$\begin{aligned}[\mathcal{I}(\theta)]_{ij} &= \mathbb{E}_{x\sim P_\theta(x)}\left[\left(\frac{1}{P_{\theta'}(x)}\right)^2\frac{\partial P_{\theta'}(x)}{\partial\theta'_i}\frac{\partial P_{\theta'}(x)}{\partial\theta'_j}\Bigg|_{\theta'=\theta}\right] \\ &= \int_{-\infty}^{\infty}P_\theta(x)\left[\frac{1}{P_\theta(x)^2}\frac{\partial P_{\theta'}(x)}{\partial\theta'_i}\frac{\partial P_{\theta'}(x)}{\partial\theta'_j}\Bigg|_{\theta'=\theta}\right]dx \quad (**)\end{aligned}$$

We know that the integral of the second derivative of the probability density is zero:

$$\int_{-\infty}^{\infty}\frac{\partial^2 P_{\theta'}(x)}{\partial\theta'_i\partial\theta'_j}\Bigg|_{\theta'=\theta}dx = \frac{\partial^2}{\partial\theta'_i\partial\theta'_j}\left(\int_{-\infty}^{\infty}P_{\theta'}(x)dx\right)\Bigg|_{\theta'=\theta} = 0$$

Subtracting this zero-valued integral from $(**)$ yields:

$$\begin{aligned}[\mathcal{I}(\theta)]_{ij} &= \int_{-\infty}^{\infty}P_\theta(x)\left[\frac{1}{P_\theta(x)^2}\frac{\partial P_{\theta'}(x)}{\partial\theta'_i}\frac{\partial P_{\theta'}(x)}{\partial\theta'_j} - \frac{1}{P_\theta(x)}\frac{\partial^2 P_{\theta'}(x)}{\partial\theta'_i\partial\theta'_j}\Bigg|_{\theta'=\theta}\right]dx \\ &= \int_{-\infty}^{\infty}P_\theta(x)\left[-\frac{\partial}{\partial\theta'_j}\left(\frac{1}{P_{\theta'}(x)}\frac{\partial P_{\theta'}(x)}{\partial\theta'_i}\right)\Bigg|_{\theta'=\theta}\right]dx \\ &= \int_{-\infty}^{\infty}P_\theta(x)\left[-\frac{\partial^2\log P_{\theta'}(x)}{\partial\theta'_i\partial\theta'_j}\Bigg|_{\theta'=\theta}\right]dx \\ &= \mathbb{E}_{x\sim P_\theta(x)}\left[-\nabla^2_{\theta'}\log P_{\theta'}(x)\Big|_{\theta'=\theta}\right]_{ij}\end{aligned}$$

Thus, $\mathcal{I}(\theta) = -\mathbb{E}_{x\sim P_\theta(x)}\left[\nabla^2_{\theta'}\log P_{\theta'}(x)\Big|_{\theta'=\theta}\right]$.

Step 3: Final Substitution. Substitute the results from Step 1 and Step 2 back into the Taylor expansion for B:

$$\begin{aligned}B &\approx A + 0 + \frac{1}{2}d^T\mathbb{E}_{x\sim P_\theta(x)}\left[\nabla^2_{\theta'}\log P_{\theta'}(x)\Big|_{\theta'=\theta}\right]d \\ &= A - \frac{1}{2}d^T\mathcal{I}(\theta)d\end{aligned}$$

Finally, substitute B back into the KL divergence equation:

$$D_{\mathrm{KL}}(P_\theta||P_{\theta+d}) \approx A - \left(A - \frac{1}{2}d^T\mathcal{I}(\theta)d\right) = \frac{1}{2}d^T\mathcal{I}(\theta)d$$

(b) We need to show that the direction of update of Newton's Method, and the direction of natural gradient, are exactly the same for GLMs.

Update rule for Newton's method:

$$\theta := \theta - H^{-1}\nabla_\theta \ell(\theta)$$

where $H_{ij} = \frac{\partial^2 \ell(\theta)}{\partial\theta_i \partial\theta_j}$ is the Empirical Hessian.

Exponential family:

$$p(y;\eta) = b(y)\exp(\eta^T T(y) - a(\eta))$$

For the sake of convenience, assume $\eta \in \mathbb{R}$ and $T(y) = y$. This makes the exponential family take the form:

$$p(y;\eta) = b(y)\exp(\eta y - a(\eta))$$

Log likelihood on GLM models:

$$\begin{aligned}\ell(\theta) &= \log \prod_{i=1}^{m} p(y^{(i)}|x^{(i)};\theta) = \sum_{i=1}^{m} \log p(y^{(i)}|x^{(i)};\theta) \\ &= \sum_{i=1}^{m} \left[\log b(y^{(i)}) + \theta^T x^{(i)} y^{(i)} - a(\theta^T x^{(i)})\right]\end{aligned}$$

Taking the first derivative:

$$\begin{aligned}\implies \nabla_\theta \ell(\theta) &= \sum_{i=1}^{m} \nabla_\theta \left[\log b(y^{(i)}) + \theta^T x^{(i)} y^{(i)} - a(\theta^T x^{(i)})\right] \\ &= \sum_{i=1}^{m} x^{(i)} \left[y^{(i)} - a'(\theta^T x^{(i)})\right]\end{aligned}$$

Taking the second derivative:

$$\implies \nabla_\theta^2 \ell(\theta) = \sum_{i=1}^{m} -a''(\theta^T x^{(i)}) x^{(i)} (x^{(i)})^T$$

$\implies \mathbb{E}_{y\sim p(y)}[H] = H$ since H is constant wrt y.

Now consider the update direction of the Natural gradient: $\mathcal{I}^{-1}(\theta)\nabla_\theta \ell(\theta)$.

We have:

$$\mathcal{I}(\theta) = \mathbb{E}_{y\sim p(y)}[-H] = -H$$

Compared to the Newton update, we see the update direction between the two methods matches for GLMs.

Problem 2.

Sandwich Function

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function, and $g : \mathbb{R}^n \to \mathbb{R}$ be a concave function. Suppose $g(\vec{x}) \leq f(\vec{x})$ for all $\vec{x} \in \mathbb{R}^n$. Prove that there exists an affine function $h : \mathbb{R}^n \to \mathbb{R}$ such that $g(\vec{x}) \leq h(\vec{x}) \leq f(\vec{x})$ for all $\vec{x} \in \mathbb{R}^n$.

Source: UCB EECS127 FA23 - Disc 8

Solution:

Since f is convex and g is concave, we have int epi(f) and hypo(g) are two nonempty disjoint convex sets:

$$\text{int epi}(f) = \{(\vec{x}, t) \mid \vec{x} \in \mathbb{R}^n, t > f(\vec{x})\}$$

$$\text{hypo}(g) = \{(\vec{x}, t) \mid \vec{x} \in \mathbb{R}^n, t \leq g(\vec{x})\}$$

By the separating hyperplane theorem, there exists a hyperplane that separates int epi(f) and hypo(g). This means there exist $\vec{a}, b, c$ such that:

$$\vec{a}^T\vec{x} + bt \geq c, \quad \forall(\vec{x}, t) \in \text{int epi}(f)$$

$$\vec{a}^T\vec{x} + bt \leq c, \quad \forall(\vec{x}, t) \in \text{hypo}(g)$$

The separating hyperplane is $(H) : \vec{a}^T\vec{x} + bt = c$.

- Suppose $b = 0$. We have:

  $$\vec{a}^T\vec{x} \geq c, \quad \forall(\vec{x}, t) \in \text{int epi}(f)$$

  $$\vec{a}^T\vec{x} \leq c, \quad \forall(\vec{x}, t) \in \text{hypo}(g)$$

  This implies $\vec{a}^T\vec{x} = c, \forall\vec{x} \in \mathbb{R}^n$, which is false since $\vec{a} \neq \vec{0}$.

- If $b < 0$, we have $(\vec{x}, \infty) \in$ int epi(f), which implies $\vec{a}^T\vec{x} + bt \geq c$ but the left hand side tends to $-\infty$.

Thus, $b > 0$. Consider $h : \mathbb{R}^n \to \mathbb{R}, h(\vec{x}) = \frac{c}{b} - \frac{1}{b}\vec{a}^T\vec{x}$.
Let $\vec{x} \in \mathbb{R}^n$ be arbitrary. Consider $(\vec{x}, g(\vec{x})) \in$ hypo(g). We have:

$$\vec{a}^T\vec{x} + bg(\vec{x}) \leq c \implies g(\vec{x}) \leq \frac{c}{b} - \frac{1}{b}\vec{a}^T\vec{x} = h(\vec{x})$$

Consider $(\vec{x}, t) \in$ int epi(f):

$$\vec{a}^T\vec{x} + bt \geq c \implies t \geq h(\vec{x}), \forall t > f(\vec{x})$$

By the ε-principle, $f(\vec{x}) \geq h(\vec{x})$.
Therefore, $h(\vec{x})$ as defined is the desired affine function. $\square$

Problem 3.

In this question we will see how combinatorial optimization problems can sometimes be solved via related convex optimization problems. Consider the problem of deciding which subset of items to sell from among a collection of n items. Selling the item i results in revenue $s_i > 0$ and a transaction cost $c_i > 0$. There is also a fixed overall cost, which we normalize to 1, which is incurred irrespective of whether any items are sold. We are interested in maximizing the *margin*, i.e. the ratio of the total revenue to the total cost (sum of the total transaction cost and the fixed overall cost). Note that this is a combinatorial optimization problem, because the set of choices is a discrete set (here it is the set of all subsets of $\{1, \ldots, n\}$). In this question we will see that it is possible to pose this combinatorial optimization problem as a convex optimization problem.

(a) Show that the original combinatorial optimization problem can be formulated as

$$\max_{\vec{x} \in \{0,1\}^n} f(\vec{x})$$

where $f(\vec{x}) := \frac{\vec{s}^\top \vec{x}}{1+\vec{c}^\top \vec{x}}$, Here $\vec{s} \in \mathbb{R}^n$ is the column vector of revenues and $\vec{c} \in \mathbb{R}^n$ is the column vector of transaction costs associated to the individual items.

(b) Show that the combinatorial optimization problem admits a convex optimization problem formulation in the sense that the convex optimization problem

$$\begin{aligned} \min_t \quad & t \\ \text{subject to:} \quad & t \geq \vec{\mathbf{1}}^\top (\vec{s} - t\vec{c})_+, \\ & t \geq 0, \end{aligned}$$

has the same value as that of the original combinatorial optimization problem. Here $\vec{z}_+$, for a vector $\vec{z} \in \mathbb{R}^n$, denotes the vector with components $\max(0, z_i)$, $i = 1, \ldots, n$. Also $\vec{\mathbf{1}}$ denotes the all-ones column vector of length n.

Note: You should also justify why the problem formulated in this part of the question is a convex optimization problem.

Hint: For given $t \geq 0$, express the condition "$f(\vec{x}) \leq t$ for every $\vec{x} \in \{0,1\}^n$" in simple terms. Note that this is related to the idea of introducing slack variables, as in Sec. 8.3.4.4 of the textbook of Calafiore and El Ghaoui.

(c) Show that the inequality constraint $t \geq \vec{\mathbf{1}}^\top (\vec{s} - t\vec{c})_+$ is active at the optimum in the convex optimization problem, namely, it will have to be satisfied with equality.

(d) How can you recover an optimal solution $\vec{x}^*$ for the original combinatorial optimization problem from an optimal value t^* for the above convex optimization problem?

Source: UCB EECS127 FA23 - Disc 8

Solution:

(a) Maximizing the margin:

$$\max_{S \in \mathcal{P}(\{1,\ldots,n\})} \frac{\text{total revenue of selling items in subset } S}{1 + \text{total transaction cost of selling items in subset } S} \tag{1}$$

Represents the subset S as a binary mask in $\mathbb{R}^n$, we rewrite (1):

$$\max_{\vec{x} \in \{0,1\}^n} f(\vec{x}), \quad f(\vec{x}) = \frac{\vec{s}^\top \vec{x}}{1 + \vec{c}^\top \vec{x}}$$

Here $\vec{s} \in \mathbb{R}^n$ is the column vector of revenues and $\vec{c} \in \mathbb{R}^n$ is the column vector of transaction costs.

(b) The problem $\max_{\vec{x} \in \{0,1\}^n} f(\vec{x})$ is equivalent to the problem:

$$\begin{aligned} \min_t \quad & t \\ \text{subject to} \quad & t \geq f(\vec{x}) \text{ for every } \vec{x} \in \{0,1\}^n \end{aligned}$$

Rewrite $t \geq f(\vec{x})$ for every $\vec{x} \in \{0,1\}^n$:

$$\begin{aligned} t \geq \frac{\vec{s}^\top \vec{x}}{1 + \vec{c}^\top \vec{x}}, \ \forall \vec{x} \in \{0,1\}^n &\Leftrightarrow t + \vec{x}^\top (t\vec{c}) \geq \vec{s}^\top \vec{x}, \ \forall \vec{x} \in \{0,1\}^n \\ &\Leftrightarrow t \geq \vec{x}^\top (\vec{s} - t\vec{c}), \ \forall \vec{x} \in \{0,1\}^n \\ &\Leftrightarrow t \geq \max_{\vec{x} \in \{0,1\}^n} \vec{x}^\top (\vec{s} - t\vec{c}) \\ &\Leftrightarrow t \geq \mathbf{1}^\top (\vec{s} - t\vec{c})_+ \end{aligned}$$

Therefore,

$$\begin{aligned} \max_{\vec{x} \in \{0,1\}^n} f(\vec{x}) = \ & \min t \\ & \text{s.t: } t \geq \mathbf{1}^\top (\vec{s} - t\vec{c})_+ \end{aligned}$$

The reformulated problem is a convex optimization problem since:

- The objective function $f_0(t) = t$ is convex in t
- The constraint $f_1(t) = -t + \max_{\vec{x} \in \{0,1\}^n} \vec{x}^\top (\vec{s} - t\vec{c}) \leq 0$ has f_1 be convex in t since:

$$\begin{aligned} f_1(t) &= -t + \max_{\vec{x} \in \{0,1\}^n} \vec{x}^\top (\vec{s} - t\vec{c}) \\ &= -t + \sum_{i=1}^n \max(0, s_i - tc_i) \end{aligned}$$

Apply Nonnegative weighted sum and pointwise maximization rule $\Rightarrow f_1$ convex.

(c) Let t^* be the solution to the optimization problem. Suppose the inequality constraint is not active at the optimum.

We have:

$$\begin{aligned}
& t^* > \mathbf{1}^\top (\vec{s} - t^* \vec{c})_+ \\
\Rightarrow \quad & t^* > \max_{\vec{x} \in \{0,1\}^n} \vec{x}^\top (\vec{s} - t^* \vec{c}) \\
\Rightarrow \quad & t^* > \vec{x}^\top (\vec{s} - t^* \vec{c}), \quad \forall \vec{x} \in \{0,1\}^n \\
\Rightarrow \quad & t^* > f(\vec{x}), \quad \forall \vec{x} \in \{0,1\}^n \\
\Rightarrow \quad & t^* > \max_{\vec{x} \in \{0,1\}^n} f(\vec{x})
\end{aligned}$$

Which contradicts the fact that $t^* = \max_{\vec{x} \in \{0,1\}^n} f(\vec{x})$.

(d) Constraint is active at optimum:

$$\begin{aligned}
\Rightarrow \quad & t^* = \max_{\vec{x} \in \{0,1\}^n} \vec{x}^\top (\vec{s} - t^* \vec{c}) \\
\Rightarrow \quad & \exists \vec{x}^* \in \{0,1\}^n : t^* = \vec{x}^{*\top} (\vec{s} - t^* \vec{c}) \\
\Rightarrow \quad & \vec{x}^* = \operatorname*{argmax}_{\vec{x} \in \{0,1\}^n} f(\vec{x})
\end{aligned}$$

Since $\vec{x}^* = \operatorname{argmax}_{\vec{x} \in \{0,1\}^n} \vec{x}^\top (\vec{s} - t^* \vec{c})$, construct $\vec{x}$ such that:

$$x_i = \begin{cases} 1 & \text{if } s_i > t^* c_i \\ 0 & \text{if } s_i \leq t^* c_i \end{cases}$$

Problem 4.

Consider an MDP with finite state and action spaces, and discount factor $\gamma < 1$. Let B be the Bellman update operator with V a vector of values for each state. I.e., if $V' = B(V)$, then

$$V'(s) = R(s) + \gamma \max_{a \in A} \sum_{s' \in S} P_{sa}(s')V(s').$$

(a) Prove that, for any two finite-valued vectors V_1, V_2, it holds true that

$$||B(V_1) - B(V_2)||_\infty \leq \gamma ||V_1 - V_2||_\infty.$$

where

$$||V||_\infty = \max_{s \in S} |V(s)|.$$

(This shows that the Bellman update operator is a "γ-contraction in the max-norm.")

(b) We say that V is a **fixed point** of B if $B(V) = V$. Using the fact that the Bellman update operator is a γ-contraction in the max-norm, prove that B has at most one fixed point—i.e., that there is at most one solution to the Bellman equations. You may assume that B has at least one fixed point.

(c) Show that in implementing value iteration, we can stop iterating once we have found a vector V such that $||B(V) - V||_\infty <$ tolerance for some predefined tolerance.

Source: CS229 Stanford 2018 - Problem set 4

Solution:

(a) Let V_1, V_2 be any two finite-valued vectors.

$$B(V_1)(s) = V_1'(s) = R(s) + \gamma \max_{a \in A} \sum_{s' \in S} P_{sa}(s')V_1(s')$$
$$B(V_2)(s) = V_2'(s) = R(s) + \gamma \max_{a \in A} \sum_{s' \in S} P_{sa}(s')V_2(s')$$

$$\implies |B(V_1)(s) - B(V_2)(s)| = \gamma \left| \max_{a \in A} \sum_{s' \in S} P_{sa}(s')V_1(s') - \max_{a \in A} \sum_{s' \in S} P_{sa}(s')V_2(s') \right|$$

$$\implies \quad ||B(V_1) - B(V_2)||_\infty = \gamma \max_{s \in S} \left| \max_{a \in A} \mathbb{E}_{s' \sim P_{sa}}[V_1(s')] - \max_{a \in A} \mathbb{E}_{s' \sim P_{sa}}[V_2(s')] \right|$$

$$\begin{aligned} \implies \quad ||B(V_1) - B(V_2)||_\infty &\overset{(1)}{\leq} \gamma \max_{s \in S} \max_{a \in A} |\mathbb{E}_{s' \sim P_{sa}}[V_1(s') - V_2(s')]| \\ &= \gamma \max_{s \in S} |\mathbb{E}_{s' \sim P_{sa^*}}[V_1(s') - V_2(s')]| \\ &\qquad \left(\text{where } a^* = \operatorname*{argmax}_{a \in A} \ |\mathbb{E}_{s' \sim P_{sa}}[V_1(s') - V_2(s')]| \right) \end{aligned}$$

$$
\begin{aligned}
&\overset{(2)}{\leq} \gamma \max_{s \in S} \mathbb{E}_{s' \sim P_{sa^*}} |V_1(s') - V_2(s')| \\
&\leq \gamma \max_{s \in S} \max_{s' \in S} |V_1(s') - V_2(s')| \\
&= \gamma \max_{s' \in S} |V_1(s') - V_2(s')| \\
&= \gamma ||V_1 - V_2||_\infty
\end{aligned}
$$

The inequality (1) follows by the fact that:

$$
|\max_x f(x) - \max_x g(x)| \leq \max_x |f(x) - g(x)|
$$

The inequality (2) follows by Jensen's inequality.

(b) Suppose B has two different fixed points V_1, V_2. From part (a), we know that B is a γ-contraction in the max-norm. Since V_1 and V_2 are different, we know $||V_1 - V_2||_\infty > 0$. We have:

$$
\begin{aligned}
& ||B(V_1) - B(V_2)||_\infty \leq \gamma ||V_1 - V_2||_\infty \\
\implies \quad & ||V_1 - V_2||_\infty \leq \gamma ||V_1 - V_2||_\infty \\
\implies \quad & 1 \leq \gamma
\end{aligned}
$$

Which contradicts the fact that $\gamma < 1$.

Thus, B has exactly one fixed point if we assume B has at least one fixed point. I have proved in the simplified setting where we consider $f : \mathbb{R} \to \mathbb{R}$ be differentiable and $f'(x) < L$, $L < 1$ that f has at least one fixed point and, in fact, exactly one. (Real Analysis note, problem 8)

Remark: Convergence of Value Iteration Algorithm

Let V^* be the fixed point of B, meaning $B(V^*) = V^*$. We have

$$
||B^t(V) - V^*||_\infty \leq \gamma^t ||V - V^*||_\infty, \quad \forall t \in \mathbb{N}
$$

This means that value iteration converges geometrically to the optimal value function V^*. That is, after t iterations, the distance between V and V^* is at most $\gamma^t ||V - V^*||_\infty$.

(c) Suppose we want $\|V - V^*\|_\infty < \epsilon$, for any $\epsilon > 0$.

By the Triangle Inequality, we have:

$$
\|V - B(V)\|_\infty + \|B(V) - V^*\|_\infty \geq \|V - V^*\|_\infty \tag{2}
$$

Since B is a γ-contraction mapping (and V^* is its fixed point, so $B(V^*) = V^*$), we know:

$$\|B(V) - V^*\|_\infty \leq \gamma \|V - V^*\|_\infty$$

Plugging this into (2), we have:

$$\begin{aligned}
& \|V - V^*\|_\infty \leq \|V - B(V)\|_\infty + \gamma \|V - V^*\|_\infty \\
\implies \quad & \|V - V^*\|_\infty - \gamma \|V - V^*\|_\infty \leq \|V - B(V)\|_\infty \\
\implies \quad & \|V - V^*\|_\infty \leq \frac{\|V - B(V)\|_\infty}{1 - \gamma} = \frac{\text{Tolerance}}{1 - \gamma}
\end{aligned}$$

Thus, we can define Tolerance $< \epsilon(1 - \gamma)$ to guarantee that $\|V - V^*\|_\infty < \epsilon$.

Problem 5.

Consider the problem of imitation learning within a discrete MDP with horizon T and an expert policy π^*. We gather expert demonstrations from π^* and fit an imitation policy π_θ to these trajectories so that

$$\mathbb{E}_{p_{\pi^*}(s)} \pi_\theta(a \neq \pi^*(s) \mid s) = \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}_{p_{\pi^*}(s_t)} \pi_\theta(a_t \neq \pi^*(s_t) \mid s_t) \leq \varepsilon,$$

i.e., the expected likelihood that the learned policy π_θ disagrees with the expert π^* within the training distribution p_{π^*} of states drawn from random expert trajectories is at most ε.
For convenience, the notation $p_\pi(s_t)$ indicates the state distribution under π at time step t while $p(s)$ indicates the state marginal of π across time steps, unless indicated otherwise.

1. Show that $\sum_{s_t} |p_{\pi_\theta}(s_t) - p_{\pi^*}(s_t)| \leq 2T\varepsilon$.

2. Consider the expected return of the learned policy π_θ for a state-dependent reward $r(s_t)$, where we assume the reward is bounded with $|r(s_t)| \leq R_{\max}$:

$$J(\pi) = \sum_{t=1}^{T} \mathbb{E}_{p_\pi(s_t)} r(s_t).$$

   (a) Show that $J(\pi^*) - J(\pi_\theta) = \mathcal{O}(T\varepsilon)$ when the reward only depends on the last state, i.e., $r(s_t) = 0$ for all $t < T$.

   (b) Show that $J(\pi^*) - J(\pi_\theta) = \mathcal{O}(T^2\varepsilon)$ for an arbitrary reward.

Source: UCB CS285 FA23 - Homework 1

Solution:

1. Let M be the event of making at least one mistake up to time step t (not including). We have:

$$p_{\pi_\theta}(s_t) = (1 - P(M))p_{\pi^*}(s_t) + P(M)p_{\text{mistake}}(s_t)$$

$$\begin{aligned}
\implies \sum_{s_t} |p_{\pi_\theta}(s_t) - p_{\pi^*}(s_t)| &= \sum_{s_t} |P(M)(p_{\text{mistake}}(s_t) - p_{\pi^*}(s_t))| \\
&= P(M) \sum_{s_t} |p_{\text{mistake}}(s_t) - p_{\pi^*}(s_t)| \\
&\leq 2P(M) \qquad (1)
\end{aligned}$$

Let m_k be the event that the first mistake happens at time step k, for $k < t$. We have:

$$M = \bigcup_{k=1}^{t-1} m_k$$

and m_i are disjoint $\implies P(M) = \sum_{k=1}^{t-1} P(m_k)$.

We also have:

$$P(m_k) = \sum_{s_k} p_{\pi^*}(s_k)\pi_\theta(a \neq \pi^*(s_k) \mid s_k) = \mathbb{E}_{p_{\pi^*}(s_k)}\pi_\theta(a \neq \pi^*(s_k) \mid s_k)$$

$$\begin{aligned}
\implies P(M) &= \sum_{k=1}^{t-1} P(m_k) \\
&= \sum_{k=1}^{t-1} \mathbb{E}_{p_{\pi^*}(s_k)}\pi_\theta(a \neq \pi^*(s_k) \mid s_k) \\
&\leq \sum_{k=1}^{T} \mathbb{E}_{p_{\pi^*}(s_k)}\pi_\theta(a \neq \pi^*(s_k) \mid s_k) \\
&\leq T\varepsilon
\end{aligned}$$

Plug the bound $P(M) \leq T\varepsilon$ into (1), we complete the proof.

2. $J(\pi) = \sum_{t=1}^{T} E_{P_\pi(s_t)} r(s_t) \quad , \quad |r(s_t)| \leq R_{\max}$

   a) Reward only depends on the last state, $r(s_t) = 0$ for all $t < T$

$$
\begin{aligned}
J(\pi^*) - J(\pi_\theta) &= \sum_{t=1}^{T} \Big(E_{P_{\pi^*}(s_t)} r(s_t) - E_{P_{\pi_\theta}(s_t)} r(s_t) \Big) \\
&= E_{P_{\pi^*}(s_T)} r(s_T) - E_{P_{\pi_\theta}(s_T)} r(s_T) \\
&= \sum_{s_T} P_{\pi^*}(s_T) r(s_T) - \sum_{s_T} P_{\pi_\theta}(s_T) r(s_T) \\
&= \sum_{s_T} (P_{\pi^*}(s_T) - P_{\pi_\theta}(s_T)) r(s_T) \\
&\leq 2T\epsilon R_{\max} = O(T\epsilon)
\end{aligned}
$$

b) For arbitrary reward

$$
J(\pi^*) - J(\pi_\theta) = \sum_{t=1}^{T} \Big(E_{P_{\pi^*}(s_t)} r(s_t) - E_{P_{\pi_\theta}(s_t)} r(s_t) \Big)
$$

Fix t, we have: $E_{P_{\pi^*}(s_t)} r(s_t) - E_{P_{\pi_\theta}(s_t)} r(s_t) = O(T\epsilon)$ (from (a))

We sum over T terms, each term is $O(T\epsilon) \Rightarrow J(\pi^*) - J(\pi_\theta) = O(T^2\epsilon)$

So we have achieved a bound on the difference in performance (measured by expected return) between optimal policy and imitation policy.